

(*) Numeric Functions:- Performs calculations on Numbers or manipulate numeric values.

1. ABS() $\Rightarrow$ Returns the absolute value of a number.

Syntax: `ABS(number);`

Example: `SELECT ABS(-25);`

Output: 25

2. CEIL() $\Rightarrow$ Returns the smallest integer greater than or equal to a number.

Syntax: `CEIL(number);`

Example: `SELECT CEIL(12.1);`

Output: 13

3. FLOOR() $\Rightarrow$ Returns the largest integer less than / equal to number.

Syntax: `FLOOR(number);`

Example: `SELECT FLOOR(12.9);` Output: 12

4. ROUND() $\Rightarrow$ Rounds a number to specified decimal places.
Syntax: `ROUND(number, decimals);`
Example: `SELECT ROUND(12.5671, 2);` Output: 12.57.
5. TRUNCATE() $\Rightarrow$ Truncates a number to specified decimal places.
Syntax: `TRUNCATE(number, decimals);`
Example: `SELECT TRUNCATE(12.5671, 2);` Output: 12.56
6. MOD() $\Rightarrow$ Returns the remainder after division.
Syntax: `MOD(number1, number2);`
Example: `SELECT MOD(10, 3);` Output: 1
7. POWER() $\Rightarrow$ Returns the number raised to a power.
Syntax: `POWER(number, power);`
Example: `SELECT POWER(2, 3);` Output: 8.
8. SQRT() $\Rightarrow$ Returns the square root.
Syntax: `SQRT(number);`
Example: `SELECT SQRT(25);` Output: 5.
- (*) Aggregate Functions:- This function perform calculations on multiple rows and return a single result.
1. COUNT() $\Rightarrow$ Counts the number of rows/values.
Syntax: `COUNT(column);`
Example: `SELECT COUNT(*) FROM employees;`
2. SUM() $\Rightarrow$ Calculates the total.
Syntax: `SUM(column);`
Example: `SELECT SUM(salary) FROM employees;`
3. AVG() $\Rightarrow$ Calculates the average. Syntax: `AVG(column);`
Example: `SELECT AVG(salary) FROM employees;`

4. MAX() $\Rightarrow$ Returns the highest value. Syntax: `MAX(column);`
Example: `SELECT MAX(salary) FROM employees;`

5. MIN() $\Rightarrow$ Returns the lowest value. Syntax: `MIN(column);`
Example: `SELECT MIN(salary) FROM employees;`

(*) Char/String Functions : Manipulate text/string values.

1. LENGTH() $\Rightarrow$ Returns the number of character/bytes in a string.

Syntax: `LENGTH(string);`

Example: `SELECT ename, LENGTH(ename) FROM emp;`

`SELECT LENGTH('ASHWIN');` Output: 6.

2. ASCII() $\Rightarrow$ Returns the ASCII value of the first character of a string.

Syntax: `ASCII(string);`

Example: `SELECT ASCII('Ammu');`

Output: 65. Because the ASCII value of A is 65.

3. UPPER() $\Rightarrow$ Converts all characters in a string to uppercase.

Syntax: `UPPER(string);`

Example: `SELECT UPPER('Rochi');` Output: ROCHI.

4. LOWER() $\Rightarrow$ Converts all characters in a string to lowercase.

Syntax: `LOWER(string);`

Example: `SELECT LOWER('Rochi');` Output: Rochi.

5. SUBSTR() $\Rightarrow$ Extracts a specified portion of a string.

Syntax: `SUBSTR(string, start-position, length);`

Example: `SELECT ename, SUBSTR(ename, 2, 8) FROM emp;`

Example 2: `SELECT SUBSTR('DATABASE', 2, 4);` Output: ATAB.

6. REPLACE() $\Rightarrow$ Replaces a specified character or substring with another value.

Syntax: `REPLACE(string, old_string,`

Example: `SELECT REPLACE('Rachitha', 'o', 'a');`

Output: Rachitha.

7. REVERSE() $\Rightarrow$ Reverses the characters in a string.

Syntax: `REVERSE(string);`

Example: `SELECT REVERSE('Tom Jerry');`

Output: yreerJ noT.

8. INSTR() $\Rightarrow$ Returns the position of the first occurrence of a substring within a string. It returns 0 if the substring is not found.

Syntax: `INSTR(string, substring);`

Example: `SELECT INSTR('TOM&JERRY', 'J');` Output: 5

(*) DATE FUNCTIONS:- Work with dates & times.

1. CURRENT_DATE() $\Rightarrow$ Returns the current date according to the MySQL server's current date.

Syntax: `CURRENT_DATE();`

Example: `SELECT CURRENT_DATE();` Output: 2026-10-02.

2. SYSDATE() $\Rightarrow$ Returns the current date & time.

Syntax: `SYSDATE();`

Example: `SELECT SYSDATE();` Output: 2026-10-02 17:30:25.

3. NOW() $\Rightarrow$ Returns the current date & time.

Syntax: `NOW();`

Example: `SELECT NOW();` Output: 2026-10-02 17:30:25.

NOTE: NOW() and SYSDATE() can look similar, but they have differences in how the current time is evaluated during statement execution.

4. YEAR() $\Rightarrow$ Extracts the year from a date or datetime value
Syntax: YEAR(date);
Example: SELECT YEAR(NOW()); Output: 2025

(*) Conversion / Date Conversion Functions :- Convert one datatype into another.

1. DATE-FORMAT $\Rightarrow$ Converts a date into a specified display format.

Syntax: DATE-FORMAT(date, format);

Example: SELECT hiredate, DATE-FORMAT(hiredate, '%m-%d-%y') FROM emp;

If hiredate is: 2025-04-30

The Output becomes: 04-30-2025.

Common format specifiers:

- %d - Day.
- %m - Month.
- %M - Month name.
- %y - 2-digit year.
- %Y - 4-digit year.

2. DATE-ADD $\Rightarrow$ Adds a specified time interval to a date.

Syntax: DATE-ADD(date, INTERVAL value unit);

Example 1: SELECT DATE-ADD(CURRENT-DATE(), INTERVAL 100 DAY);

If Today's date is:

2026-10-02.

Output: 2027-01-10

Example 2: SELECT DATE-ADD('2026-10-02', INTERVAL 10 DAY);

Output: 2026-10-12.

3. DATE-SUB() $\Rightarrow$ Subtracts a specified time interval from a date.

Syntax: `DATE-SUB(date, INTERVAL value unit);`

Example: `SELECT DATE-SUB(CURDATE(), INTERVAL 100 DAY);`

If today's date is 2026-10-2, Output: 2026-06-24-

4. STR-TO-DATE() $\Rightarrow$ Converts a string into a DATE value according to the specified format.

Syntax: `STR-TO-DATE(string, format);`

Example: `SELECT STR-TO-DATE('30-04-2025', '%d-%m-%Y')`

Output: 2025-04-30